

# 4.5 Project Structure Overview — CampusCare

CampusCare is organized as a **monorepo** with two independent applications sharing the same repository:

|               |                                                        |
|---------------|--------------------------------------------------------|
| CampusCare/   |                                                        |
| ■■■ backend/  | Express.js REST API (TypeScript + Prisma + PostgreSQL) |
| ■■■ frontend/ | React Native mobile app (Expo SDK 54)                  |
| ■■■ docs/     | Project documentation                                  |

## Backend Structure

|                                     |                                                                                     |
|-------------------------------------|-------------------------------------------------------------------------------------|
| backend/                            |                                                                                     |
| ■■■ prisma/                         |                                                                                     |
| ■ ■■■ schema.prisma                 | Database schema and ORM model definitions                                           |
| ■ ■■■ seed.ts                       | Script to populate the database with test accounts and sample data                  |
| ■ ■■■ setup.sql                     | Raw SQL reference used during initial Supabase setup                                |
| ■                                   |                                                                                     |
| ■■■ src/                            |                                                                                     |
| ■ ■■■ server.ts                     | Entry point – creates the Express app, registers all middleware and routes, starts  |
| ■ ■                                 |                                                                                     |
| ■ ■■■ config/                       |                                                                                     |
| ■ ■ ■■■ db.ts                       | Exports the shared Prisma client instance used across all controllers               |
| ■ ■ ■■■ swagger.ts                  | Swagger/OpenAPI spec configuration for the interactive API docs at /api/docs        |
| ■ ■                                 |                                                                                     |
| ■ ■■■ controllers/                  | Business logic – one file per feature area                                          |
| ■ ■ ■■■ auth.controller.ts          | Register, login, logout, get current user (JWT issuance)                            |
| ■ ■ ■■■ issues.controller.ts        | Create, list, get, update status, assign, close, comment, upload photo              |
| ■ ■ ■■■ notifications.controller.ts | List notifications, mark all as read                                                |
| ■ ■ ■■■ admin.controller.ts         | List all users, toggle active status, grant/revoke roles, list worker               |
| ■ ■                                 |                                                                                     |
| ■ ■■■ routes/                       | Express routers – map HTTP methods and URLs to controller functions                 |
| ■ ■ ■■■ auth.routes.ts              | /api/auth/*                                                                         |
| ■ ■ ■■■ issues.routes.ts            | /api/issues/*                                                                       |
| ■ ■ ■■■ notifications.routes.ts     | /api/notifications/*                                                                |
| ■ ■ ■■■ manager.routes.ts           | /api/manager/* (MANAGER + ADMIN only)                                               |
| ■ ■ ■■■ admin.routes.ts             | /api/admin/* (ADMIN only)                                                           |
| ■ ■                                 |                                                                                     |
| ■ ■■■ middleware/                   | Reusable Express middleware                                                         |
| ■ ■ ■■■ auth.ts                     | JWT verification (requireAuth) and role-based access guard (requireRole)            |
| ■ ■ ■■■ error.ts                    | Global error handler – catches unhandled errors and returns consistent JSON error   |
| ■ ■ ■■■ upload.ts                   | Multer configuration for handling multipart photo uploads; saves files to uploads/  |
| ■ ■                                 |                                                                                     |
| ■ ■■■ services/                     |                                                                                     |
| ■ ■ ■■■ notify.ts                   | Helper function that creates Notification records in the database for a given user  |
| ■ ■                                 |                                                                                     |
| ■ ■■■ validators/                   |                                                                                     |
| ■ ■ ■■■ schemas.ts                  | Zod validation schemas for all request bodies (register, login, issue creation, sta |
| ■                                   |                                                                                     |
| ■■■ uploads/                        | Runtime folder where uploaded issue photos are stored and served as static files    |
| ■■■ .env                            | Environment variables (DATABASE_URL, JWT_SECRET, PORT – not committed to git)       |
| ■■■ .env.example                    | Template showing required environment variable keys                                 |
| ■■■ package.json                    | npm scripts and dependency declarations                                             |
| ■■■ tsconfig.json                   | TypeScript compiler configuration                                                   |

## Key Files Explained

# Frontend Structure

(Full frontend structure omitted from preview here but included in generated PDF)

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## Data Flow Summary

